

Experiment No. 9

Design and Implement flipflops using xilinx vivado

Aim:

To design and Implement flipflops using Verilog HDL and Verify their operation.

Apparatus required.

- ↳ xilinx Vivado
- ↳ System

Theory:

- ↳ Open xilinx Vivado and Create a new project
- ↳ Write the Verilog Code for the required flipflop
- ↳ Create a test bench
- ↳ Compile the design and check the Syntax errors.
- ↳ Simulate the design
- ↳ Verify the Output waveform

Programme:

(i) SR flipflop

SR- flipflop.

```
module sr_ff (clk, s, r, q);
```

```
input clk, s, r;
```

```
output reg q;
```

```
always @(posedge clk)
```

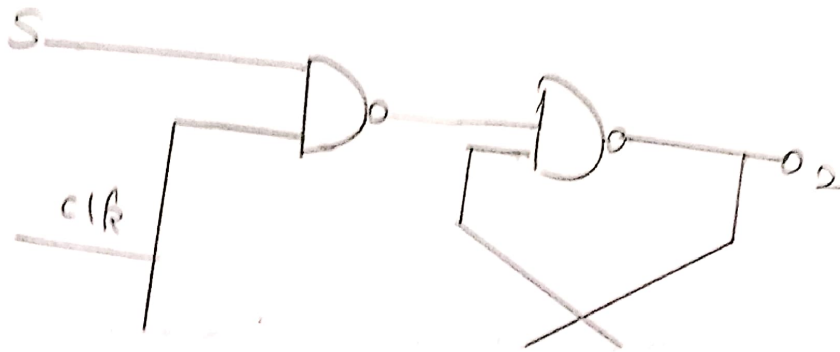
```
begin
```

```
if (s == 1'b0 & r == 1'b0)
```

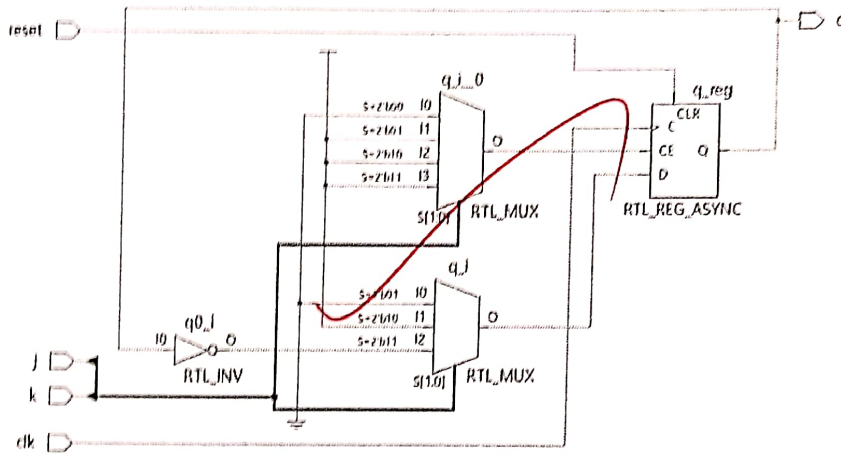
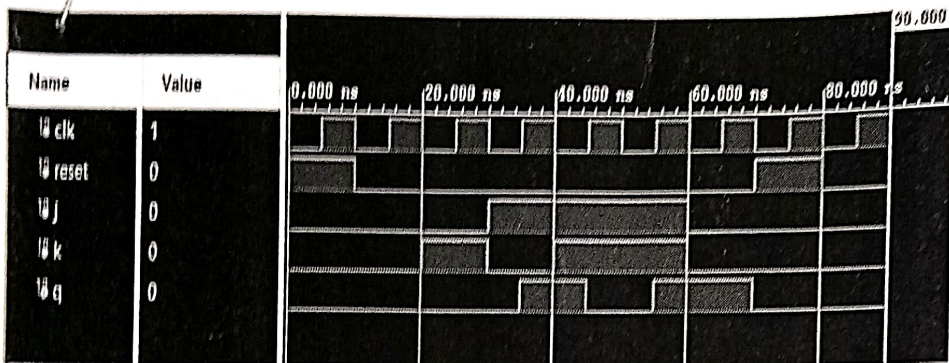
AP
verified

Diagram/ Logical & Truth table:

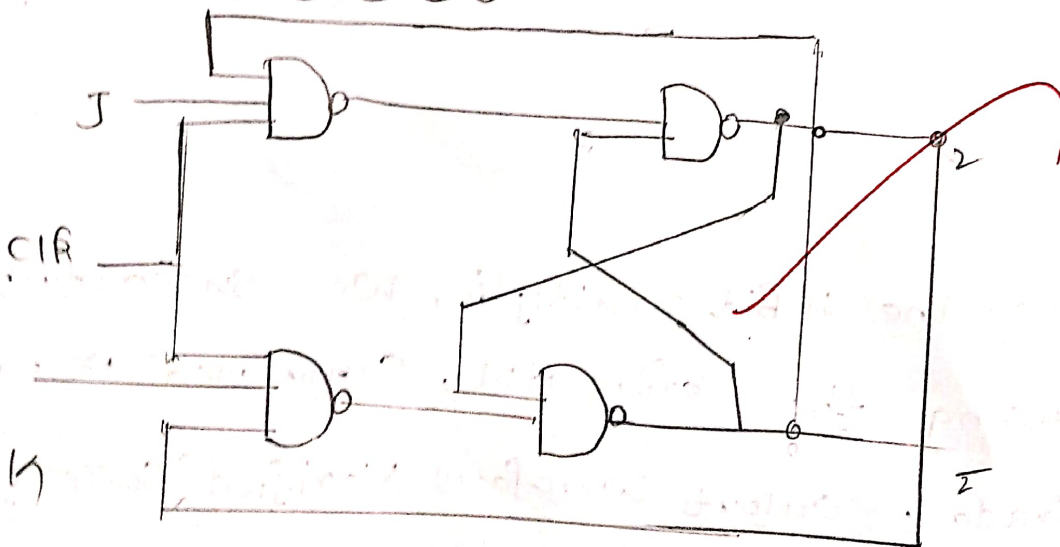
(i) SR flip flop:



JK FLIP FLOP

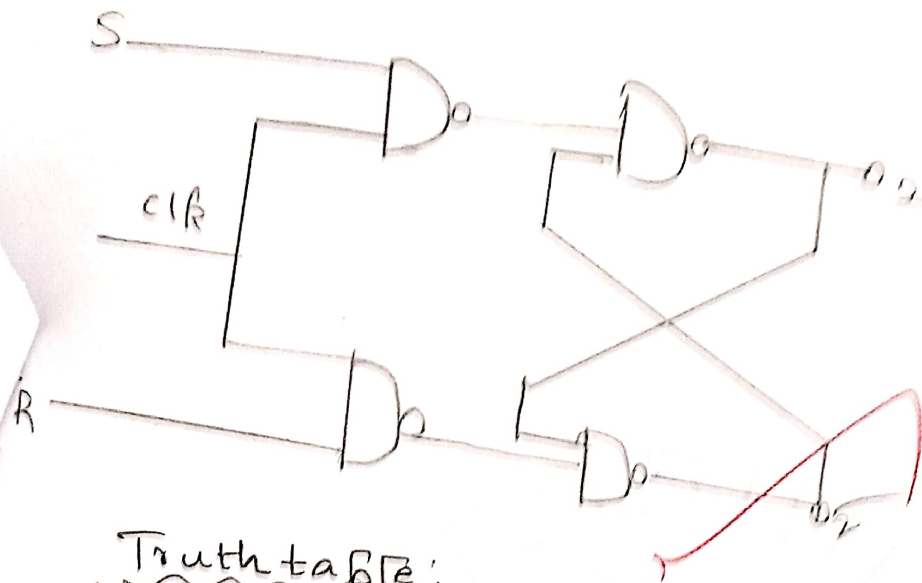


(ii) JK - flip flop:



Diagram/ Logical & Truth table

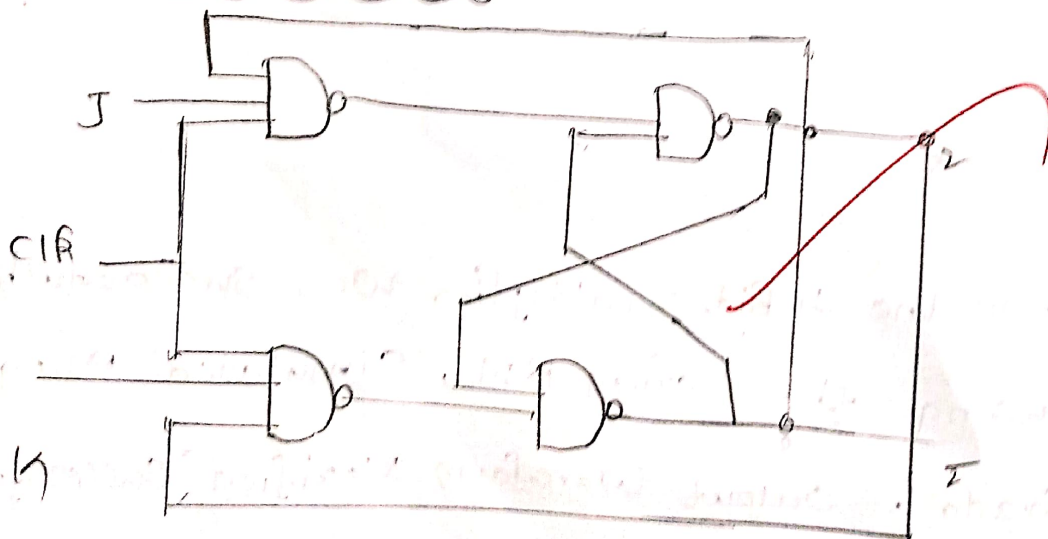
(i) SR flip flop:



Truth table:

S	R	Q_N	Q_{N+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

(ii) JK - flip flop:



```

q = q'
else if (s == 1'b1 & r == 1'b1)
q <= 1'b0;
else if (s == 1'b1 & r == 1'b0)
else
q <= 1'b1;
end
endmodule

```

(ii) JK flip flop

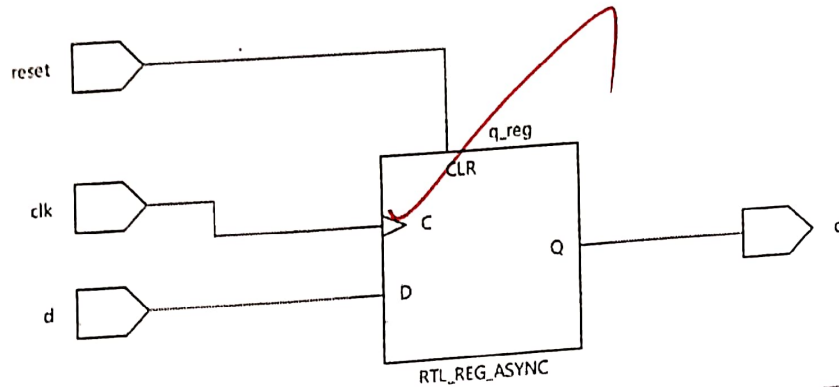
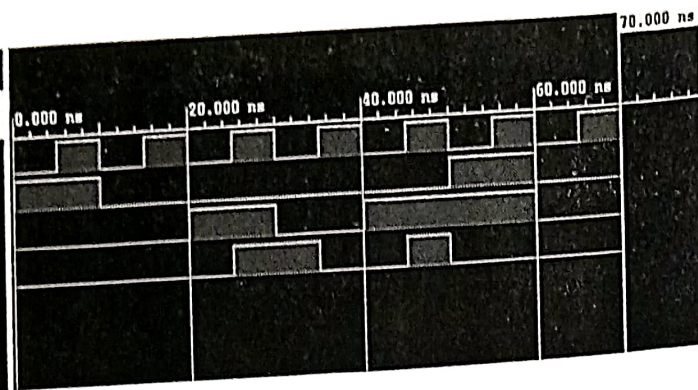
```

module jk_ff (clk, j, k, q)
input (clk, j, k)
output reg q;
always @ (posedge clk)
begin
if (j == 1'b0 & k == 1'b0)
q <= q;
else if (j == 1'b0 & k == 1'b1)
q <= 1'b0;
else if (j == 1'b1 & k == 1'b0)
q <= 1'b1;
else
q <= ~q;
end
endmodule

```

o/p reg

Name	Value
clk	1
reset	0
d	0
q	0



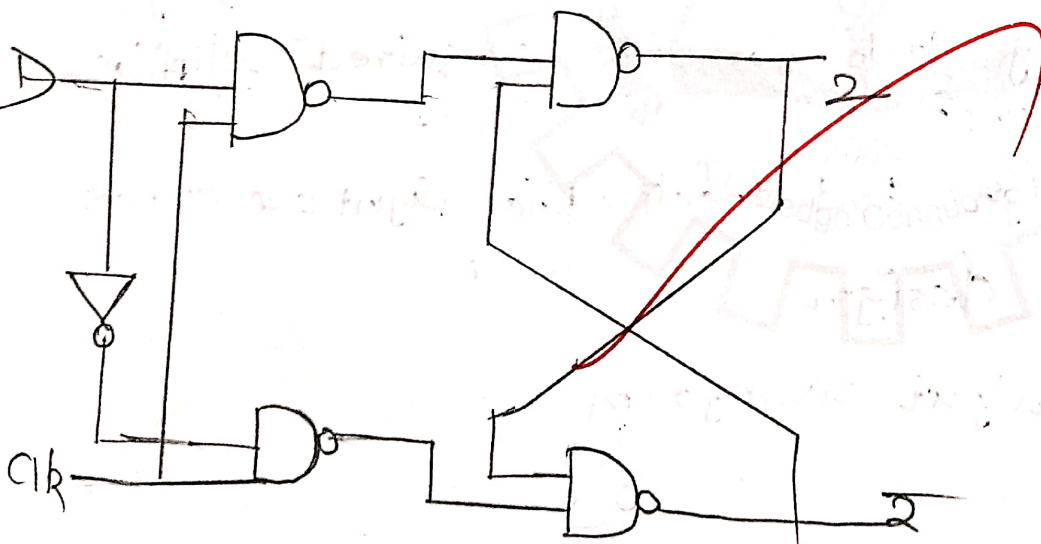
a	D	$Q_i(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

a	D	$Q_i(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

Truth table

J	K	Q_N	Q_{N+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

D flip flop



Truth table

Q	D	$Q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

(iii) D flipflop

```
module d-ff (clk, d, q);  
  input clk, d;  
  output reg q;  
  always @(posedge clk)  
  begin  
    q <= d;  
  end  
endmodule.
```

*DP
works*

(iv) T flipflop

```
module t-ff (clk, t, q);  
  input clk, t;  
  output reg q;  
  always @(posedge clk)  
  begin  
    if (t == 1'b0)  
      q <= q;  
    else  
      q <= ~q;  
    end  
  end  
endmodule.
```

*DP
works*

Result:

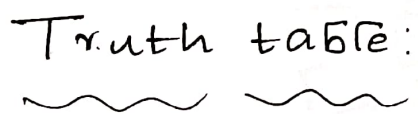
Thus the SR, JK, D and T flipflop were successfully designed using Verilog HDL simulated xilinx vivado, and the output verified successfully.

*DP
works*

T FLIP FLOP

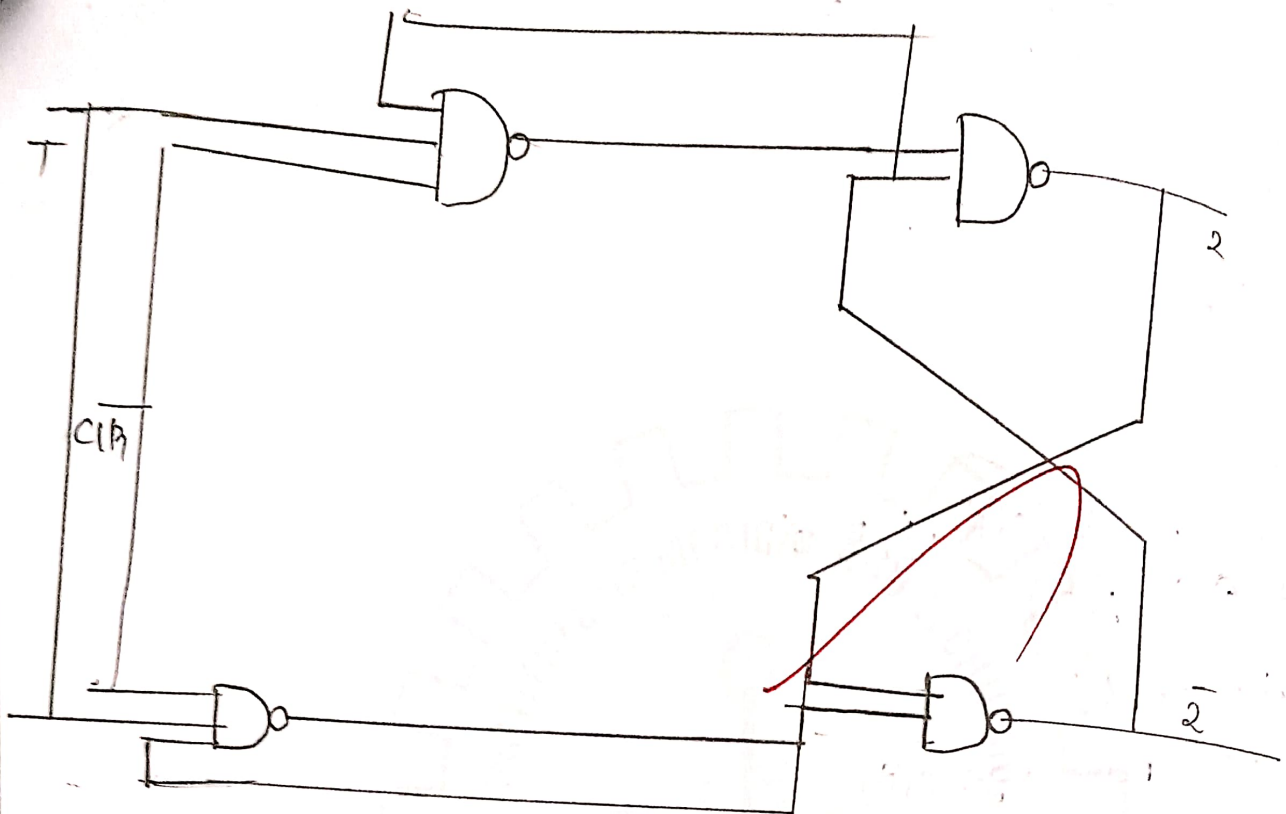
Name	Value
clk	1
reset	0
t	0
q	0

0.000 ns 20.000 ns 40.000 ns



T	q_N	q_{N+1}
0	0	0
0	1	1
1	0	1
1	1	0

T flip flop:



Truth table:

T	Q_N	Q_{N+1}
0	0	0
0	1	1
1	0	1
1	1	0